

API	Instruction			Operand	Description
2703	D	JOG		S ₁ , S ₂ , S ₃ , D ₁ , D ₂	JOG output

Device	X	Y	M	S	T	C	HC	D	FR	SM	SR	E	K	16#	"\$"	F
S_1					●	●	●	●	●		○		○	○		
S_2					●	●	●	●	●		○		○	○		
S_3					●	●	●	●	●		○		○	○		
D_1		○														
D_2		○	○													

Data type	BOOL	WORD	DWORD	LWORD	UINT	INT	DINT	LINT	REAL	LREAL	TMR	CNT	STRING
S_1		●	●			●	●						
S_2		●	●			●	●						
S_3		●	●			●	●						
D_1	●												
D_2	●												

Pulse Instruction	16-bit instruction	32-bit instruction
—	AS	AS

Symbol

JOG			DJOG		
En			En		
S_1		D_1	S_1		D_1
S_2		D_2	S_2		D_2
S_3			S_3		

S_1 : Ramp-up time
 S_2 : Target output frequency
 S_3 : Ramp-down time
 D_1 : Pulse output device
 D_2 : Auxiliary output device

Explanation

- This instruction specifies the JOG output (S_1 , S_2 , S_3) for the devices specified in D_1 and D_2 . S_1 is the ramp-up time with the unit of 10 ms. For Example, the value 10 means the ramp-up time is 100 ms. After the instruction is executed, the output frequency can be divided by ten, and it speeds up once every 10ms. It reaches the JOG target frequency specified by S_2 at the ramp-up time S_1 .
- When the instruction is disabled, its output frequency ramps down once every 10ms based on the ramp-down time in S_3 . The output does not stop until reaching the ramp-down time. If you set the ramp-down time to 0, the output stops immediately.
- The range of the ramp-up time S_1 and the ramp-down time S_3 is between 0–3000 (0–30 seconds). If the setting value is out of range, the PLC processes it as the minimum or maximum output value. The ramp-up time and ramp-down time are affected by the scan time. If you require accurate ramp-up or ramp-down time, it is recommended that you use an output instruction with specified ramp up or ramp down time such as the DDRVI instruction (API 2706).
- The range of the target output frequency S_2 is between -200 kHz to 200 kHz. If the setting value is out of the range, the PLC processes it as the minimum or maximum output value. A positive number for the output frequency indicates forward output. A negative number for the output frequency indicates reverse output.
- D_1 allows only Y0.0–Y0.11 for the output point. For the auxiliary output point specified by D_2 , refer to the output points in the following table. If you choose other output points or the M device, the value in SR indicating the output mode is invalid and the Pulse+direction mode is valid by default. D_2 is a direction output point.

D_1 selects the even output point number.

Output point for D_1	Y0.0	Y0.2	Y0.4	Y0.6	Y0.8	Y0.10
Direction output point for D_2	Y0.1	Y0.3	Y0.5	Y0.7	Y0.9	Y0.11
Busy flag	SM460	SM480	SM500	SM520	SM540	SM560
Output mode	SR462	SR482	SR502	SR522	SR542	SR562
Present output position	SR460	SR480	SR500	SR520	SR540	SR560
	SR461	SR481	SR501	SR521	SR541	SR561
Backlash compensation	SR478	SR498	SR518	SR538	SR558	SR578

D_1 selects the odd output point number.

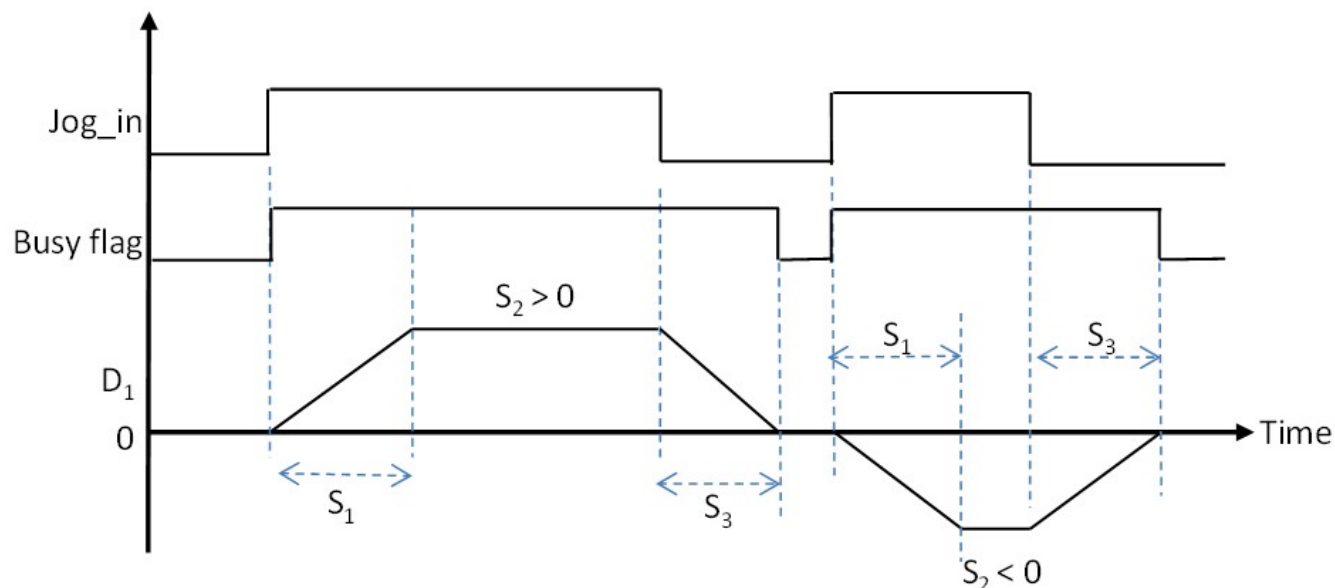
Output point for D_1	Y0.1	Y0.3	Y0.5	Y0.7	Y0.9	Y0.11
Direction output point for D_2	Y1.0 or above or any M device (BOOL)					
Busy flag	SM472	SM492	SM512	SM532	SM552	SM572
Present output position	SR474	SR494	SR514	SR534	SR554	SR574
	SR475	SR495	SR515	SR535	SR555	SR575
Backlash compensation	SR479	SR499	SR519	SR539	SR559	SR579

6. When the value is 0 in SR462, SR482, SR502, SR522, SR542, and SR562, it indicates the Pulse+direction output mode. When the value is 1 in SR462, SR482, SR502, SR522, SR542, and SR562, it indicates the A/B phase output mode.
Note: you can select the output mode only when D_1 uses an even output point and D_2 uses a recommended direction output point.
7. When the direction outputs are not using the default outputs to output, you can refer to the table below for setting up the outputting time that pulse outputting goes first before direction outputting does. So that you can be sure the outputting only happens when switching to the direction outputs. The unit for SR is millisecond and the range is 0-20 ms. Default is 0, indicating inactive.

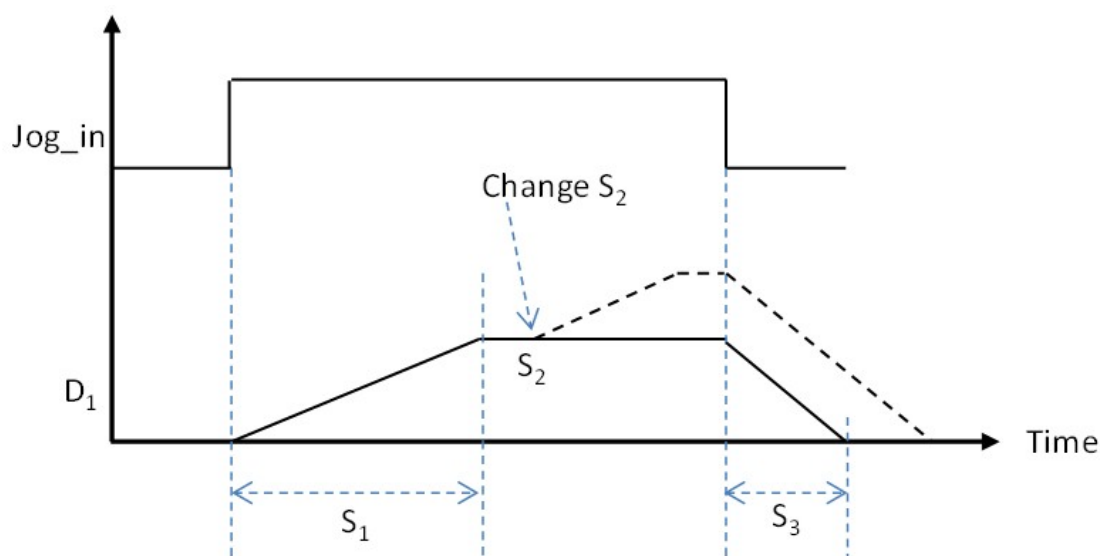
Output point for D_1	Y0.0	Y0.1	Y0.2	Y0.3	Y0.4	Y0.5
SR number	SR640	SR641	SR642	SR643	SR644	SR645
Output point for D_1	Y0.6	Y0.7	Y0.8	Y0.9	Y0.10	Y0.11
SR number	SR646	SR647	SR648	SR649	SR650	SR651

Note: the time source here is from PLC scan time. (available for firmware version 1.02.20 or later)

8. When the output begins, the Busy flag SM is ON. When the output is completed, the Busy flag is automatically reset to OFF and the Completion flag is not ON.
9. The following graph shows the output timing diagram. For the Busy flag in the following graph, refer to the Busy flag axis.



10. You can modify the target output frequency during instruction execution but you can NOT change the ramp-up or ramp-down time. When the new target frequency is greater than the previous one, the instruction uses the ramp-up slope. When the new frequency is less than the previous one, the instruction uses the ramp-down slope.
11. The instruction sets the ramp-up/down slope through conversion of the set time and target frequency when the instruction is executed. The slope does not change with the changing target frequency in the output process. For example, the original target output frequency is 1 kHz and then it is modified to 2 kHz. The actual ramp-down time is different from the original time. The dotted line is the ramp-up and ramp down timing diagram after the target frequency is modified.



12. If there is any change on the frequency during ramp up or ramp down, the change can only be carried out after the original frequency completes outputting. For example, if the new target frequency is 100 Hz and the ramp-up timing is set at 100 (1 second), after the original frequency completes outputting, the new target frequency 100 Hz will be outputted next. (Even though the target frequency changes every 10 ms but if the pulse does NOT complete outputting, the change can NOT be applied.)

Example (ST program):

It is suggested to use ladder diagrams as the programming language in this instruction. If you need to use structured texts, you need to use the output completion auto-reset flag as well for the PLC to know the output point is free and ready to be used again when the output completes. Taking Y0.0 as the output point, the program should be written as below. See the following example.

```
0001 IF M0 THEN
0002     DJOG (1000,5000,100,Y0.0,Y0.1);
0003     M1 := TRUE;
0004 ELSIF (NOT M0) AND M1 THEN
0005     SM470 := TRUE;
0006     M1 := FALSE;
0007 END IF;
```